Tochukwu Bedford

Sr. Full-Stack Software Engineer
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Full-stack engineer with 5+ years shipping performance-sensitive products used by thousands of music creators. I work across the whole stack, from Angular and React interfaces to NestJS backends, real-time data pipelines, and cloud infrastructure on AWS and Cloudflare. At Offtop I'm a top-two contributor across the web, backend, and mobile codebases, and the lead engineer on the mobile app.

Experience

Offtop (offtop.com), Sr. Full-Stack Engineer

Feb 2024 to Present
Tech: TypeScript, NestJS, Angular, RxJS, React, Electron, Posthog, React Native, MongoDB, Redis, AWS, Cloudflare

- Ship weekly production releases across Offtop's web, desktop, and mobile apps. I'm a top-two contributor across all three core codebases (roughly 2,200, 570, and 540 commits on web, backend, and mobile) and the lead engineer on the mobile app.
- Built the real-time layer that pushes live upload progress and AI audio tagging to every client, using WebSockets, Server-Sent Events, and MongoDB change streams that resume cleanly after a restart without losing events.
- I rolled out PostHog analytics across Offtop's entire stack (web, backend, mobile, and desktop) on my own, with a shared event schema so the same user actions line up the same way on every platform. It gave the team its first unified view of funnels and retention, plus session replay and feature flags, and became the data we lean on for product and growth calls.
- Designed and single-handedly built Offtop Link, a cross-platform Electron desktop app that watches local folders and syncs audio to the cloud, streaming straight to S3 with a resumable queue. Set up its signed release pipeline too: notarized macOS builds, Windows builds signed through Azure Key Vault, and auto-updates from S3.
- Built an iMessage assistant on OpenAI tool-calling (9 tools) that lets producers manage their library by texting. It spreads sends across a pool of phone numbers with daily caps so none of them get flagged as spam.
- Wrote the audio transcoding workers (ffmpeg, WAV to M4A and lossless FLAC) as a separate service fed by SQS and BullMQ, and moved lossless files from S3 to Cloudflare R2 to stop paying egress fees.
- Lead contributor on the React Native app (~540 commits since February). Built its real-time sync over Server-Sent Events for uploads, track replacement, and video export, and a low-latency audio player that starts in about a third of a second by streaming the first chunk over an HTTP range request and loading the rest in the background.
- Sped up the web app with an abortable, throttled upload engine, moved waveform rendering off the main thread onto a server endpoint, and added virtual scrolling for long libraries.

Axxess Cavista Technology, Front End Engineer

Jan 2023 to Aug 2024 · **Tech:** Aurelia, RxJS, TypeScript, SCSS

- Built interfaces used by more than 30,000 healthcare professionals, with support for older browsers down to Chrome 55 and Safari 7.

Devos, Backend Engineer (Contract)

Dec 2023 to Apr 2024 · **Tech:** Node.js, Python, SQL, AWS, OpenAPI

- Built backend APIs for a fintech identity product, covering logins, photo verification, and KYC, along with the OpenAPI specs the frontend built against.

Social Crucifixion, Software Engineer

Jan 2022 to Dec 2023 · Los Angeles, CA · **Tech:** React, Redux, Firebase

- Built landing pages that drove 200+ signups and over 1,000 visits, including 3D product views through Google Model Viewer.

Skills

- **Languages:** TypeScript, JavaScript, Node.js, Python, SQL, Sass, C
- **Frameworks and libraries:** NestJS, Angular, React, React Native, Electron, RxJS, Express, Socket.IO, Jest, Vitest
- **Data:** MongoDB, Redis, PostgreSQL, Firebase
- **Cloud and infrastructure:** AWS (Elastic Beanstalk, S3, SQS, CloudWatch), Cloudflare (Workers, R2), Pulumi, Docker, GCP, Azure Key Vault, GitHub Actions
- **Real-time and messaging:** WebSockets, Server-Sent Events, MongoDB change streams, SQS and BullMQ queues
- **Tools:** Sentry, PostHog, Git, Figma, Postman

Projects and Open Source

- **Querci** (querci.net, beta). A file-sharing app for the music industry focused on resumable, verified uploads. I handled everything from billing to keeping the server fast under load. Demo video.
- **44DB Collective** (44db.net). A site for a Nigerian producer collective, built in React and TypeScript with a Sanity CMS so they can update it themselves.
- **React Fast Marquee.** Added a mount callback the library was missing (PR #68, shipped in 1.6.0).

Education

Covenant University, B.Eng (First class Hons.), Mechanical Engineering · *2017 to 2022*